HuggingFace Transformers – 30th March

Monday, 30 March 2026 6:06 PM

NLP

Contextual text → numerical form

- One Hot encoding,
- Bag of word
- TF-IDF

} → sparse representation.

- word embedding. → dense, continuous vector.
  - Word2Vec
  - Glove
  - FastText

  } → Problems.
    - Context Ignorance.
    - Poor long-range dependency.

- RNN
  - Idea
    - process the sentence or sequence step by step.
    - maintain a hidden state. (memory of previous word).
  - Limitation
    - Vanishing Gradient problem

- LSTM (long-short term memory).
  - Idea

    Introduce gate
    - Forget gate
    - Input gate
    - Output gate.

    Advantage

    Better to handle long term dependencies RNN

    Limitation
    - Sequential. → Training is slow & inefficient
    -

- GRU (Gated Recurrent unit).

  Simplified version of LSTM.
  - Fewer gates → faster than LSTM.

  Advantage of GRU.
  - Computationally lighter
  - As equivalent performance to LSTM in many task

  Limitation
  - Sequential model.
  - Struggles with very long dependencies
  - Not scalable for large dataset

- Attention mechanism

  Instead of remembering everything, we focus on important words.

  Advantage

  - Better contextual understanding.
  - Handle long term dependencies more efficiently.

  Limitation

  - Complex architecture.
  - Sequential model.

2018 → Research paper → Attention is all you need.

- Remove recurrence completely.
- only use attention mechanism.

→ Instead of processing words one by one, look at all word at once and learn the relationship directly.

Transformers.

↳ models

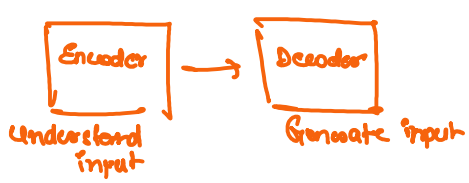
↳ revolutionized NLP.

Core Idea

- Self attention mechanism

Advantage

- Better context understanding.
- Parallel process → faster training.
- Long-range dependencies handle
- Scalable.



- Input Embedding.
  - word2vec.
  - Glove
  - Fasttext

  } input → numerical vector

- Positional encoding.
  - Use sine & cosine functions → positional vector
  - Add this positional vector to embedding.

- Self attention mechanism

Query → what I am looking for
Key → what I contain
value → Information I provide

- $\text{Attention}(Q,K,V) = \text{softmax}\left\{\frac{Q \cdot K^T}{\sqrt{d_k}}\right\} \cdot V.$

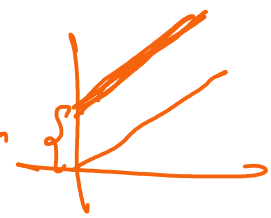
→ Steps.
Compare Query & all Key.
Get Importance score
Multiply with value
Combine result

- Multi-head attention

- Feed Forward Network
Process each word independently after attention

$y = mx + c$ → residual.

FFN → changed representation + original representation



- Layer normalization

LLM → GenAI
LIM.

Krish Naik

- Embedding.
- Positional encoding.
- Self attention
- Multi head attention
- Feed Forward
- Residual Connection
- Layer Norm